

Fall 2015

Problem 1 (Fa15). Find the real values of x for which

$$\sum_{n=0}^{\infty} \frac{(1/2)(3/2) \cdots ((2n-1)/2)}{n!} \left(\frac{2x}{1+x^2} \right)^{2n} =$$
$$= 1 + \frac{1}{2} \left(\frac{2x}{1+x^2} \right)^2 + \frac{1}{2} \frac{3}{4} \left(\frac{2x}{1+x^2} \right)^4 + \cdots$$

converges and sum it for these numbers.

Problem 2 (Fa15). Show that there is a real valued function on the real plane that is not continuous, but is continuous when restricted to any straight line.

Problem 3 (Fa15). Let $f(x)$ be differentiable on an interval (a, b) .

1. Prove that if X is the range of $(f(u)-f(v))/(u-v)$ for $a < u < v < b$ and Y is the range of $f'(x)$ on (a, b) then $X \subseteq Y \subseteq \overline{X}$.
2. Prove that the range of $f'(x)$ on (a, b) is an interval, possibly unbounded.

Problem 4 (Fa15). Let $S = \mathbb{C} \cup \{\infty\}$ be the Riemann sphere, and define the map $\varphi : \mathbb{C}^2 \setminus \{(0, 0)\} \rightarrow S$ by $\varphi(w, z) = w/z$ for $z \neq 0$, $\varphi(w, 0) = \infty$.

Prove that there is a unique map $\tau : S \rightarrow S$ with the following property: $\tau(\varphi(w, z)) = \varphi(w', z')$ if and only if the one-dimensional subspaces $\mathbb{C} \cdot (w, z)$ and $\mathbb{C} \cdot (w', z')$ are orthogonal under the standard Hermitian inner product on $\mathbb{C}^2$ in which the unit vectors $(1, 0)$ and $(0, 1)$ are orthonormal. Show that τ is continuous and bijective. Determine whether τ is holomorphic or not.

Problem 5 (Fa14, Fa15, Fa21). 1. Suppose $p(z) = c(z - a_1) \cdots (z - a_n)$ is a complex polynomial. If z has positive real part and all the zeros a_i have negative real part, show that $p'(z)/p(z)$ has positive real part.

2. Show that all the zeros of the derivative p' of a complex polynomial lie in the convex hull of the zeros of p .

Problem 6 (Fa15). In the Euclidean space $\mathbb{R}^4$, consider the “hyperellipsoid” $2x^2 + 3y^2 + 4z^2 + 5u^2 = 1$. Does there exist a 3-dimensional subspace passing through the origin which intersects the ellipsoid in a sphere?

Problem 7 (Fa15). It is a corollary to the Jordan canonical form theorem that $n \times n$ matrices in Jordan canonical form, all of whose eigenvalues are zeros, are similar if and only if the sizes of their Jordan blocks coincide, up to permutations. Prove this directly, without using the Jordan canonical form theorem.

Problem 8 (Fa15). Find all the subgroups of the dihedral group D_{12} , the group of symmetries of a regular hexagon.

Problem 9 (Fa15). Show that $x^3 - 2x$ is an injective function from the rational numbers to the rational numbers.

Problem 10 (Fa15). For a real, find a 2-dimensional space of real valued solutions of $y'' = ay/x^2$ for $x > 0$. When $a = -1/4$ find the solution with $y = 0$, $y' = 1$ at $x = 1$.

Problem 11 (Fa15, Fa24). Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a function, and assume that:

f is continuous on $[0, \infty)$;

f is differentiable on $(0, \infty)$;

$f'(x) \leq 0$ for all $x > 0$ such that $f(x) > 1$; and

$f(0) = 1$.

Prove that $f(x) \leq 1$ for all $x \geq 0$.

Problem 12 (Fa15). The unit cube in the space $\mathcal{C}_{[0,1]}$ of continuous real valued functions on the interval is defined as the subset

$$\left\{ f \in \mathcal{C}_{[0,1]} \mid \|f\| = \sup_{0 \leq t \leq 1} |f(t)| \leq 1 \right\}$$

Prove that there exists a 2-dimensional linear subspace in $\mathcal{C}_{[0,1]}$ whose intersection with the unit cube is a circular disc.

Problem 13 (Fa15). A *Schur function* is a holomorphic function defined on the open unit disc whose values have absolute value at most 1. Show that if f is a nonconstant Schur function then

$$\frac{f(0) - f(z)}{(1 - \overline{f(0)}f(z))z}$$

is also a Schur function.

Problem 14 (Fa15). Find all entire functions $f(z)$ such that $\Re(f(x+iy)) = x^3y - xy^3$. Express your answer directly in terms of z , not in terms of x and y .

Problem 15 (Fa15). Given a positive integer n , let $\dots c_{-1}, c_0, c_1, \dots$ be a sequence of real numbers with period n , that is, $c_{k+n} = c_k$ for all $k \in \mathbb{Z}$. Let C be the $n \times n$ matrix defined by $c_{ij} = c_{j-i}$. Prove that all matrices of this form, for n fixed, have a common Hermitian orthonormal basis of complex eigenvectors, find these eigenvectors, and the corresponding eigenvalues.

Problem 16 (Su79, Sp82, Sp83, Su84, Fa91, Fa98, Sp11, Fa15, Fa16, Fa23). Let $\mathbb{F}$ be a finite field with q elements and let V be an n -dimensional vector space over $\mathbb{F}$.

1. Determine the number of elements in V .
2. Let $\text{GL}_n(\mathbb{F})$ denote the group of all $n \times n$ nonsingular matrices over $\mathbb{F}$. Determine the order of $\text{GL}_n(\mathbb{F})$.
3. Let $\text{SL}_n(\mathbb{F})$ denote the subgroup of $\text{GL}_n(\mathbb{F})$ consisting of matrices with determinant 1. Find the order of $\text{SL}_n(\mathbb{F})$.

Problem 17 (Fa79, Fa87, Sp10, Fa15). Let $M_{n \times n}(\mathbb{F})$ be the ring of $n \times n$ matrices over a field $\mathbb{F}$. Prove that it has no 2-sided ideals except $M_{n \times n}(\mathbb{F})$ and $\{0\}$.

Problem 18 (Fa15). How many ways are there to arrange 8 rooks on an 8×8 chessboard so that no two attack each other (in other words, each row and column contains exactly one rook), where two ways are counted as the same if they are equivalent under one of the 8 symmetries of the chessboard?